Ahmedabad University

CSE 631: Machine Learning

Weekly Report(3)

(Group - 9) Learning Machine

Section 1

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Weekly Progress Report - 1

Machine Learning–Based Automated Stone Counting System for Embroidered Fabrics

1. Detailed Summary of Work Done During the Current Week

The third week of the project was devoted to the change of the focus towards the practical work with components of the system. The main task was the creation of a pipeline that processes the image, extracts salient visual features of the embroidery images, and the creation of the feature matrix necessary to train the machine-learning models. This step was crucial to transform raw image data to organized numerical data which can be used successfully by machine-learning algorithms.

1.1 The implementation of Image Preprocessing Pipeline can be described in the following way:

The first activity that was done during the present week was the creation of an image preprocessing module. The pictures created through embroidery often have noise, uneven lighting and intricate background patterns due to the threading and zari fabrications; therefore, preprocessing plays a central role in improving image fidelity before further analysis processes are undertaken to the image.

The preprocessing pipeline adopted includes the following steps:

- Image Loading and Resizing: The embroidery images with high resolution were loaded and resized to a standardized size in order to maintain consistency in processing.
- Noise Reduction: the Gaussian blur filtering was used to reduce the noise in the image and eliminate minor irregularities.
- Contrast Enhancement: Contrast-Limited Adaptive Histogram Equalization (CLAHE) was used to enhance the local contrast, hence decorative stones.

- Illumination normalization Basic normalization methodologies were used to reduce illumination variation across images.
- Such preprocessing steps helped a great deal to enhance the distinguishability of the regions of stones and reduce the effect of background embroidery patterns.

1.2 Gradient Based feature extraction.

The next important task was to get gradient-based features on the images after preprocessing. Gradient properties are salient in nature as decorative stones normally build discrete intensity changes in comparison with the neighbouring embroidery threads.

The gradient based methods adopted are:

- Sobel Gradient Filters: Horizontal and vertical gradient was calculated in order to identify intensity difference.
- Gradient Magnitude Calculation: Gradient strength was builded up in order to highlight edges and stone boundaries.
- Edge Enhancement: Areas that had strong gradients were highlighted as possible stone areas.
- These gradient effects help the take of structural differences between rocks and non-stone areas in the textile.

1.3 The feature extraction is based on the location of the region.

Besides the gradient values, region-based features were also obtained to outline the shape and the visualization of possible stone regions. These characteristics provide very important geometric information which helps distinguish decorative stones and embroidery threads.

The features that have been extracted include:

- Area of detected regions
- Perimeter of regions
- Rounded stones shape measure Circularity measure to detect circularity measures.
- Boundary sizes of identified elements.
- These geometrical features are also informative, since most of the decorative stones have compact and relatively circular morphologies.

1.4 The construction of the feature matrix is performed in the following step.

After extraction of features, then the next step was to transform the features into an organized data to be used in training of machine-learning models.

- The procedure encompassed:
- Pooling of all extracted features (gradient, geometric and intensity based)
- Arrangement into feature table.
- A candidate region is represented by each row.
- Every column represents a particular feature extracted.

The feature matrix is the major input in the training of classifiers like the Support Vector Machines and the random forest models.

The data on the features were saved in the structured tabular format supporting the further experimentation and analysis with the help of Pandas DataFrames.

1.5 Initial Visualization and Validation

Several visualization methods were used to ensure that the process of preprocessing and feature extraction was running correctly.

These included:

- The presentation of original and processed pictures.
- Visualizing gradient maps
- Mapping the identified candidate regions to embroidery images.

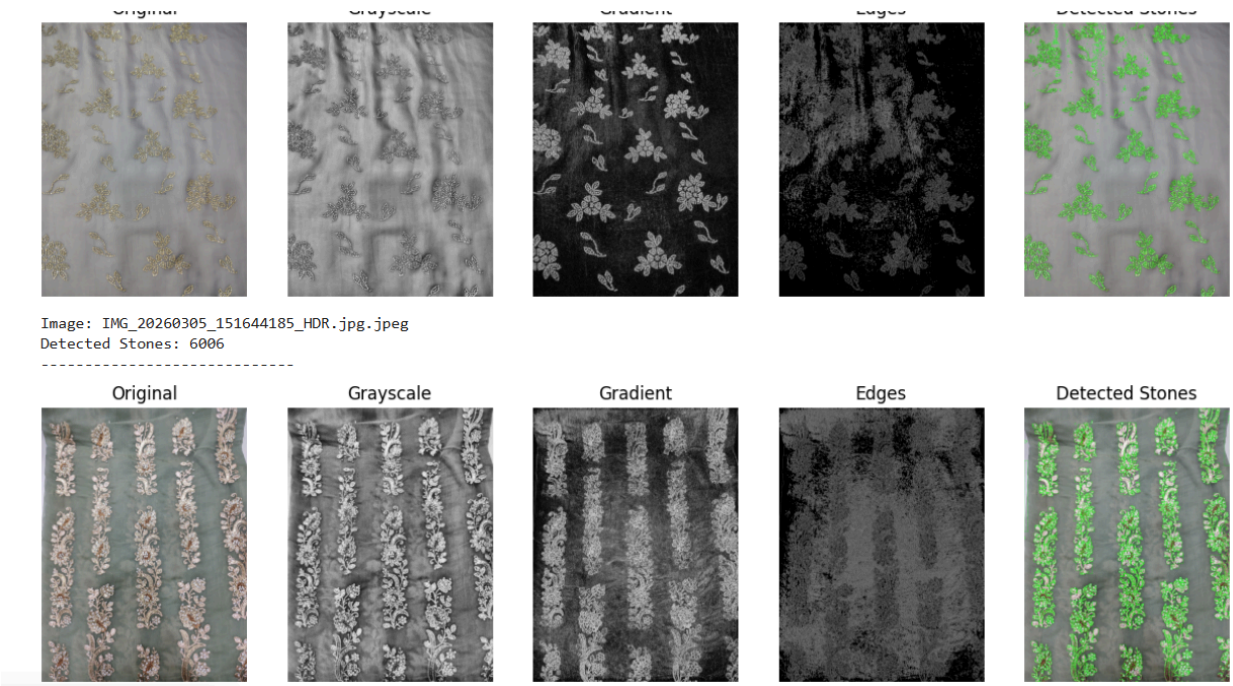
This graphical confirmation helped to make sure that stone areas have been marked accordingly and that the feature that was extracted had some meaning.

1.6 Future Development of Machine Learning Pipeline.

Since the preprocessing and feature extraction modules have been implemented successfully, the project has been introduced to a junction where the dataset necessary to train the machine-learning models is ready.

Output table created at this stage will be the input of the next stage, which will be training and testing classification models that will be used to detect stones.

Sample image:



2. Activities to be undertaken in the Week ahead.

The implementation of machine learning models and consequently, the evaluation of their performance will be dedicated to the following week, where the primary focus should be put on the accurate detection and listing of stone artifacts.

2.1 The training of the machine learning model follows as follows:

Fit traditional machine learning models using the built feature matrix:

- Support Vector Machine (SVM)
- Random Forest Classifier

The trained models will either be classified into Stone and Non-Stone candidate regions.

2.2 Model Evaluation

Test the performance of the trained models with the help of appropriate evaluation metrics including:

- Accuracy
- Precision
- Recall
- F1-score

Also, perform conduct error analysis to define the misclassified regions.

2.3 Stone Counting Implementation

This is achieved by allocating lessons to students with differing abilities, varying their activity mix, and varying their learning tools to provide suitable support for each group.<|human|>Stone Counting Implementation: This is implemented by assigning lessons to students of different abilities, a different mix of activities, and different learning tools to ensure each group receives appropriate support.

Create a module which uses the trained classifier to:

- Detect valid stone regions
- tally the number of stones that are in each embroidery picture.

This module will be the main functionality of the automated stone-counting system.

2.4 Performance Comparison

Compare the machine-learning-based solution to simple image-processing solutions to determine the enhancement of the level of detection and counting accuracy.

2.5 Analysis of Visualization and Results.

Develop plots and visualisation to demonstrate the performance results, including:

- Accuracy comparison charts
- Error distribution plots
- Sample detection visualisation.